

20260813

546 NH gvcf files

- gvcf files have been saved on my project2 space.

```
(glike) bash-4.4$ pwd
/project2/mahj_1946/MEC_WGS_NH_546_gvcf
(glike) bash-4.4$ du -sh .
1.2T
(glike) bash-4.4$ ls -a | wc -l
1628
```

- gvcf files have been saved to the Elements hard drive, still stored in my office space

[archived]MEC_WGS_NH_546_gvcf/					/project2/mahj_1946/MEC_WGS_NH_546_gvcf/				
Name	Size	Type	Changed		Name	Size	Changed	Rights	Owner
scripts		File folder	8/13/2026 4:51:31 AM		EC008168.hard-filtered.gvcf.gz.tbi	948 KB	8/10/2026 8:52:48 AM	rw-rw-r--	mahj
EC707022.hard-filtere...	1,180 KB	TBI File	8/10/2026 8:37:14 AM		EC008176.hard-filtered.gvcf.gz	1,746.43...	8/10/2026 9:24:00 AM	rw-rw-r--	mahj
EC707022.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:50:32 AM		EC008176.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:06:33 AM	rw-rw-r--	mahj
EC707022.hard-filtere...	2,209.7...	WinRAR archive	8/10/2026 8:39:07 AM		EC008176.hard-filtered.gvcf.gz.tbi	1,096 KB	8/10/2026 8:51:26 AM	rw-rw-r--	mahj
EC702807.hard-filtere...	1,228 KB	TBI File	8/10/2026 8:54:47 AM		EC008177.hard-filtered.gvcf.gz	1,443.04...	8/10/2026 9:25:44 AM	rw-rw-r--	mahj
EC702807.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:08:34 AM		EC008177.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:01:53 AM	rw-rw-r--	mahj
EC702807.hard-filtere...	2,362.9...	WinRAR archive	8/10/2026 8:46:13 AM		EC008177.hard-filtered.gvcf.gz.tbi	970 KB	8/10/2026 8:03:34 AM	rw-rw-r--	mahj
EC701813.hard-filtere...	1,047 KB	TBI File	8/10/2026 8:39:44 AM		EC008398.hard-filtered.gvcf.gz	1,283.84...	8/10/2026 8:58:34 AM	rw-rw-r--	mahj
EC701813.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:49:36 AM		EC008398.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:07:08 AM	rw-rw-r--	mahj
EC701813.hard-filtere...	1,616.0...	WinRAR archive	8/10/2026 8:41:20 AM		EC008398.hard-filtered.gvcf.gz.tbi	903 KB	8/10/2026 9:27:10 AM	rw-rw-r--	mahj
EC701686.hard-filtere...	968 KB	TBI File	8/10/2026 8:45:01 AM		EC008400.hard-filtered.gvcf.gz	2,495.90...	8/10/2026 8:54:07 AM	rw-rw-r--	mahj
EC701686.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:41:57 AM		EC008400.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:20:40 AM	rw-rw-r--	mahj
EC701686.hard-filtere...	1,465.6...	WinRAR archive	8/10/2026 9:00:26 AM		EC008400.hard-filtered.gvcf.gz.tbi	1,242 KB	8/10/2026 8:57:29 AM	rw-rw-r--	mahj
EC701657.hard-filtere...	1,253 KB	TBI File	8/10/2026 8:55:39 AM		EC008409.hard-filtered.gvcf.gz	1,561.22...	8/10/2026 8:43:07 AM	rw-rw-r--	mahj
EC701657.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:16:06 AM		EC008409.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:15:33 AM	rw-rw-r--	mahj
EC701657.hard-filtere...	2,594.0...	WinRAR archive	8/10/2026 9:03:19 AM		EC008409.hard-filtered.gvcf.gz.tbi	1,020 KB	8/10/2026 8:54:13 AM	rw-rw-r--	mahj
EC699382.hard-filtere...	1,300 KB	TBI File	8/10/2026 8:25:02 AM		EC008450.hard-filtered.gvcf.gz	2,406.93...	8/10/2026 8:39:19 AM	rw-rw-r--	mahj
EC699382.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:06:10 AM		EC008450.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 8:38:41 AM	rw-rw-r--	mahj
EC699382.hard-filtere...	3,205.3...	WinRAR archive	8/10/2026 8:43:59 AM		EC008450.hard-filtered.gvcf.gz.tbi	1,255 KB	8/10/2026 8:56:36 AM	rw-rw-r--	mahj
EC689678.hard-filtere...	1,232 KB	TBI File	8/10/2026 8:23:08 AM		EC008517.hard-filtered.gvcf.gz	2,839.96...	8/10/2026 9:20:47 AM	rw-rw-r--	mahj
EC689678.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:41:02 AM		EC008517.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 8:42:50 AM	rw-rw-r--	mahj
EC689678.hard-filtere...	2,319.0...	WinRAR archive	8/10/2026 9:04:09 AM		EC008517.hard-filtered.gvcf.gz.tbi	1,291 KB	8/10/2026 9:18:15 AM	rw-rw-r--	mahj
EC688093.hard-filtere...	1,312 KB	TBI File	8/10/2026 8:37:18 AM		EC008703.hard-filtered.gvcf.gz	1,689.36...	8/10/2026 9:09:35 AM	rw-rw-r--	mahj
EC688093.hard-filtere...	1 KB	MDSSUM File	8/10/2026 9:11:50 AM		EC008703.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:21:14 AM	rw-rw-r--	mahj
EC688093.hard-filtere...	3,970.5...	WinRAR archive	8/10/2026 9:03:55 AM		EC008703.hard-filtered.gvcf.gz.tbi	1,076 KB	8/10/2026 8:53:48 AM	rw-rw-r--	mahj
EC204636.hard-filtere...	1,159 KB	TBI File	8/10/2026 9:25:40 AM		EC008777.hard-filtered.gvcf.gz	2,904.24...	8/10/2026 8:38:22 AM	rw-rw-r--	mahj
EC204636.hard-filtere...	1 KB	MDSSUM File	8/10/2026 9:18:57 AM		EC008777.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 8:45:13 AM	rw-rw-r--	mahj
EC204636.hard-filtere...	2,011.9...	WinRAR archive	8/10/2026 8:56:29 AM		EC008777.hard-filtered.gvcf.gz.tbi	1,259 KB	8/10/2026 9:06:10 AM	rw-rw-r--	mahj
EC204623.hard-filtere...	1,162 KB	TBI File	8/10/2026 8:37:14 AM		EC008776.hard-filtered.gvcf.gz	1,864.50...	8/10/2026 8:50:41 AM	rw-rw-r--	mahj
EC204623.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:41:57 AM		EC008776.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:05:23 AM	rw-rw-r--	mahj
EC204623.hard-filtere...	2,087.9...	WinRAR archive	8/10/2026 8:50:01 AM		EC008776.hard-filtered.gvcf.gz.tbi	1,128 KB	8/10/2026 8:50:05 AM	rw-rw-r--	mahj
EC204591.hard-filtere...	1,238 KB	TBI File	8/10/2026 9:23:08 AM		EC009019.hard-filtered.gvcf.gz	2,100.07...	8/10/2026 9:04:44 AM	rw-rw-r--	mahj
EC204591.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:35:57 AM		EC009019.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 9:17:06 AM	rw-rw-r--	mahj
EC204591.hard-filtere...	2,421.7...	WinRAR archive	8/10/2026 9:15:25 AM		EC009019.hard-filtered.gvcf.gz.tbi	1,180 KB	8/10/2026 8:51:56 AM	rw-rw-r--	mahj
EC204576.hard-filtere...	1,283 KB	TBI File	8/10/2026 8:36:43 AM		EC009098.hard-filtered.gvcf.gz	1,935.22...	8/10/2026 9:00:22 AM	rw-rw-r--	mahj
EC204576.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:42:56 AM		EC009098.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 8:57:25 AM	rw-rw-r--	mahj
EC204576.hard-filtere...	2,692.0...	WinRAR archive	8/10/2026 9:21:59 AM		EC009098.hard-filtered.gvcf.gz.tbi	1,136 KB	8/10/2026 9:01:02 AM	rw-rw-r--	mahj
EC204573.hard-filtere...	1,239 KB	TBI File	8/10/2026 8:51:22 AM		EC009180.hard-filtered.gvcf.gz	1,517.26...	8/10/2026 9:14:24 AM	rw-rw-r--	mahj
EC204573.hard-filtere...	1 KB	MDSSUM File	8/10/2026 8:49:52 AM		EC009180.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 8:48:00 AM	rw-rw-r--	mahj
EC204573.hard-filtere...	2,379.6...	WinRAR archive	8/10/2026 9:13:23 AM		EC009180.hard-filtered.gvcf.gz.tbi	1,000 KB	8/10/2026 9:26:37 AM	rw-rw-r--	mahj
EC204356.hard-filtere...	1,164 KB	TBI File	8/10/2026 9:05:55 AM		EC010936.hard-filtered.gvcf.gz	2,218.63...	8/10/2026 9:12:26 AM	rw-rw-r--	mahj
EC204356.hard-filtere...					EC010936.hard-filtered.gvcf.gz.md5sum	1 KB	8/10/2026 8:55:17 AM	rw-rw-r--	mahj

gLike Benchmarking

- I've been tackling a few key questions:
 - Question 1: Does the number of trees affect the relative log likelihood of a fit, given the same set of initial condition?
 - Question 2: If given the true simulation parameters as starting parameters, how does the standard gLike optimizer handle these initial conditions?
 - Question 3 (WIP): Is CMA-ES a "better" optimizer than the standard gLike optimizer?
 - Question 4 (WIP): Does a log likelihood parameter transform improve runtime or fit?
- To tackle these questions, I set up a set of simulation / fitting jobs.
 - Varying the number of trees from 1-20

- Varying the initial conditions:
 - Using a "standard" set of initial condition (based on gLike documentation)
 - Using the true simulation parameters as initial conditions (to test for the case in which the iterator somehow finds other params that fit better)
- Simulate the standard native hawaiian model
using `msprime`
- Split genome into NUMTREES different equidistant trees
- Compute the log likelihood of the true ARG parameters --> `logp_true`
- Fit arg to the simulated output and compute the log likelihood --> `logp_x`
- Compare `logp_x` to `logp_true`
- Arbitrary initial conditions:

NUMTREE	logp_x	logp_true	runtime
1	-11595.168965764871	-11490.346261456412	4h 57m 55.0s
2	-23255.293549866048	-23014.326340593623	4h 31m 24.9s
3	-34516.00796552041	-34574.15313597546	3h 39m 20.9s
4	-46008.97851301703	-46007.92964036898	3h 43m 14.5s
5	-57602.847723188584	-57656.19177214663	4h 35m 49.1s
6	-68769.3947739363	-68910.97713724486	4h 28m 39.0s
7	-80506.19047673274	-80688.3912425454	4h 22m 55.8s
8	-91828.67668214749	-92110.04302356433	4h 20m 3.5s
9	-103599.14879110621	103852.1301750249	4h 31m 22.0s
10	-114814.55648935284	-115577.97202309304	3h 58m 8.2s
11	-126600.84073514448	-126541.75744535157	4h 25m 15.6s
12	-138248.30284521537	-138360.36357885879	4h 37m 6.3s
13	-148678.10077407598	-149699.37505869137	5h 7m 54.4s
14	-160763.51491720954	-161202.3462876855	4h 29m 26.4s
15	-172363.64902671456	-173007.39748194552	4h 15m 19.0s
16	-183788.71704638223	-184509.43559347308	4h 36m 48.6s

NUMTREE	logp_x	logp_true	runtime
17	-195266.46893355504	-195745.96172770177	4h 44m 8.9s
18	-206505.21517516958	-206987.8067480764	4h 42m 56.1s
19	-218316.4707371188	-219310.62837241986	4h 14m 39.3s
20	-229685.76989039988	-231030.16712193147	5h 0m 60.0s

- "True" initial conditions

NUMTREE	logp_x	logp_true	runtime
1	-11449.02219934836	-11487.013641243895	3h 22m 47.0s
2	-22817.965125401097	-23019.820971163772	2h 43m 49.4s
3	-34357.40037054017	-34589.323884104684	2h 56m 26.1s
4	-45758.030939156546	-46029.401700858834	4h 23m 13.9s
5	-57244.121879477774	-57663.93574300182	4h 42m 44.1s
6	-68546.38051208953	-68896.55531159393	5h 53m 22.7s
7	-80117.3847675181	-80712.22903118923	4h 41m 19.2s
8	-91507.69726252931	-92043.75753323105	3h 22m 16.9s
9	-103011.81141177715	103851.37363198126	2h 58m 36.1s
10	-114468.37141922988	-115571.47408603117	2h 57m 31.6s
11	-125805.14560235296	-126579.7406975803	2h 13m 40.1s
12	-137364.3694068794	-138343.64480433546	2h 2m 50.2s
13	-160116.1081812764	-161245.37486025316	2h 55m 24.0s
14	-171685.67278993476	-172970.7618297966	3h 37m 58.5s
15	-183052.11607500492	-184489.28985116474	2h 44m 38.0s
16	-183052.11607500492	-184489.28985116474	2h 44m 38.0s
17	-194504.4042610448	-195744.34164408693	3h 23m 12.8s
18	-205776.92119260645	-207008.1376598309	2h 46m 58.1s

NUMTREE	logp_x	logp_true	runtime
19	-217534.00346765874	-219357.3235833446	3h 28m 50.5s
20	-22817.965125401097	-23019.820971163772	3h 22m 47.0s

snpArcher vs. CCDG Methodology comparison

- Mirchandani CD, Shultz AJ, Thomas GWC, Smith SJ, Baylis M, Arnold B, Corbett-Detig R, Enbody E, Sackton TB. A Fast, Reproducible, High-throughput Variant Calling Workflow for Population Genomics. Mol Biol Evol. 2024 Jan 3;41(1):msad270. doi: 10.1093/molbev/msad270. PMID: 38069903; PMCID: PMC10764099.
 - a. snpArcher is a similar workflow to the GATK joint-calling approach
 - b. It incorporates GATK HaplotypeCaller (per-sample variant discovery), GenomicsDBImport (consolidation of gVCFs), and then GenotypeGVCFs (cohort-wide joint genotyping)
 - c. Major contribution is automation / reproducibility of GATK workflow, particularly usable with HPC clusters
- CCDG Methodology
 - a. GATK HaplotypeCaller (per-sample variant discovery)
 - b. GVCf outputs were concatenated into complete per-sample GVCfs using **Picard MergeVcfs v2.4.1**.
 - c. GATK **ReblockGVCfs (Poplin et al. 2017)** was used to reduce GVCf size
 - Poplin R, Ruano-Rubio V, DePristo MA, Fennell TJ, Carneiro MO, Van der Auwera GA, Kling DE, Gauthier LD, Levy-Moonshine A, Roazen D, et al. 2018. Scaling accurate genetic variant discovery to tens of thousands of samples. bioRxiv 201178. <https://doi.org/10.1101/201178>, preprint: not peer reviewed.
 - d. Reblocked GVCfs were combined into multi-sample Hail VariantDataset (VDS) files using VariantDatasetCombiner
 - Hail Team. Hail 0.2.13-81ab564db2b4. <https://github.com/hail-is/hail/releases/tag/0.2.13>.
 - e. Joint genotyping with GnarlyGenotyper
 - f. Variant-quality annotation using VQSR
 - DePristo M, Banks E, Poplin R, Garimella K, Maguire J, Hartl C, Philippakis A, del Angel G, Rivas MA, Hanna M, McKenna A, Fennell T, Kernytzky A, Sivachenko A, Cibulskis K, Gabriel S, Altshuler D, Daly M. (2011). A framework for variation discovery and genotyping using next-generation DNA sequencing data. Nat Genet, 43:491-498. DOI: 10.1038/ng.806.
 - g. VDS files were converted to Hail MatrixTable (MT) format, with multi-allelic variants decomposing into bi-allelic variants

h. Some quality control metrics (samples removed for abnormal heterozygosity, indels, consent withdrawal, relatedness, etc.)

- Comparison
 - CCDG is customized for an extremely large human WGS dataset (ReblockGVCF for scalability, QC for scalability, VDS to MT file format change for scalability, etc.)
 - CCDG input slightly differs from `snpArcher` -- CCDG uses sequencing data while `snpArcher` can use `FASTQs` or SRA accessions while automatically obtaining the reference
 - Both frameworks use GATK HaplotypeCaller for variant discovery
 - Both frameworks use BWA-MEM; with CCDG using Picard MarkDuplicates to concatenate / remove duplicates and `snpArcher` marking duplicates as part of the automated workflow
 - CCDG QC is specific to human WGS, `snpArcher` is more generalizable to different organisms
 - CCDG uses Hail VDS + GnarlyGenotyper for joint-genotyping while `snpArcher` uses a more standard GATK GenomicsDBImport + GenotypeGVCFs
 - CCDG Uses VDS + Gnarlygenotyper for parallelization as well; `snpArcher` uses Snakemake
 - CCDG Output is highly QC'd WGS multi-sample callset in Hail MatrixTable format, while `snpArcher` produces a multi-sample VCF for downstream analysis.
 - **In General**, CCDG is highly customized specifically for extremely large human WGS datasets, while `snpArcher` is a generalizable, easily-implemented framework that uses standard approaches

TODO

- Figure out how to release data on `dbgap` (mostly paperwork), Terra --> `dbgap`